
Estimating the Causal Treatment Effect of Job Training Programs on Participant Earnings

Evidence from the U.S.

22.06.2026

Lena Fiedler

Github: https://github.com/lefie1231/CausalInference_ML

Content

Look at Data &
Structural Model

Extract Treatment Effects

Inspect
Heterogeneity

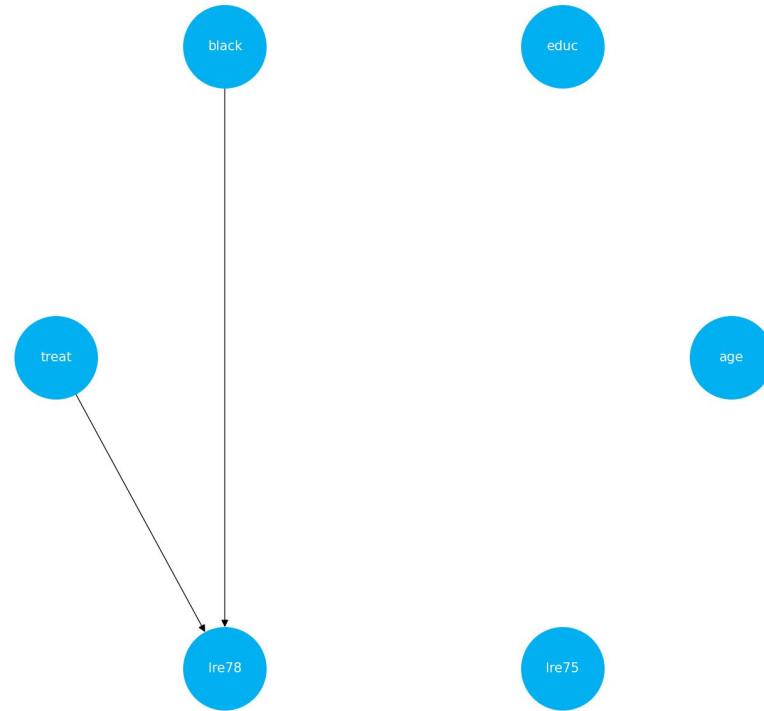
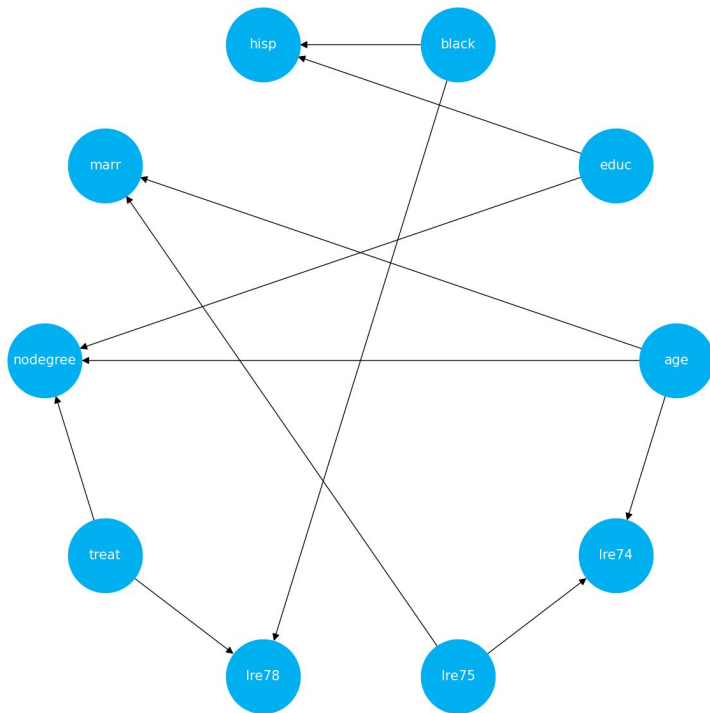
Does participation in a job training
program increase real earnings for
disadvantaged people?

Which group benefits the
most?

Data

	treat	age	educ	black	hisp	marr	nodegree	re74	re75	re78
count	445	445	445	445	445	445	445	445	445	445
mean	0,42	25,37	10,20	0,83	0,09	0,17	0,78	2.102,27	1.377,14	5.300,76
std	0,49	7,10	1,79	0,37	0,28	0,37	0,41	5.363,58	3.150,96	6.631,50
min	0	17	3	0	0	0	0	0	0	0
25%	0	20	9	1	0	0	1	0	0	0
50%	0	24	10	1	0	0	1	0	0	3.701,81
75%	1	28	11	1	0	0	1	824,39	1.220,84	8.124,71
max	1	55	16	1	1	1	1	39.570,68	25.142,24	60.307,93

Causal Structural Model

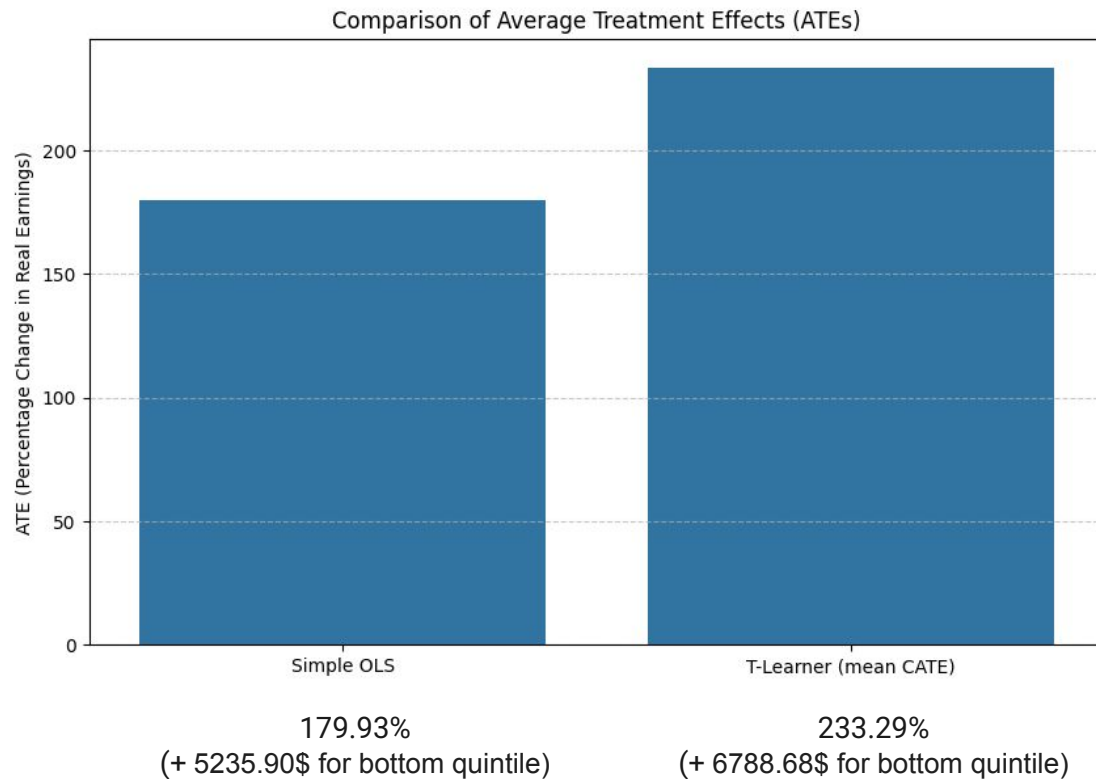


Model Selection

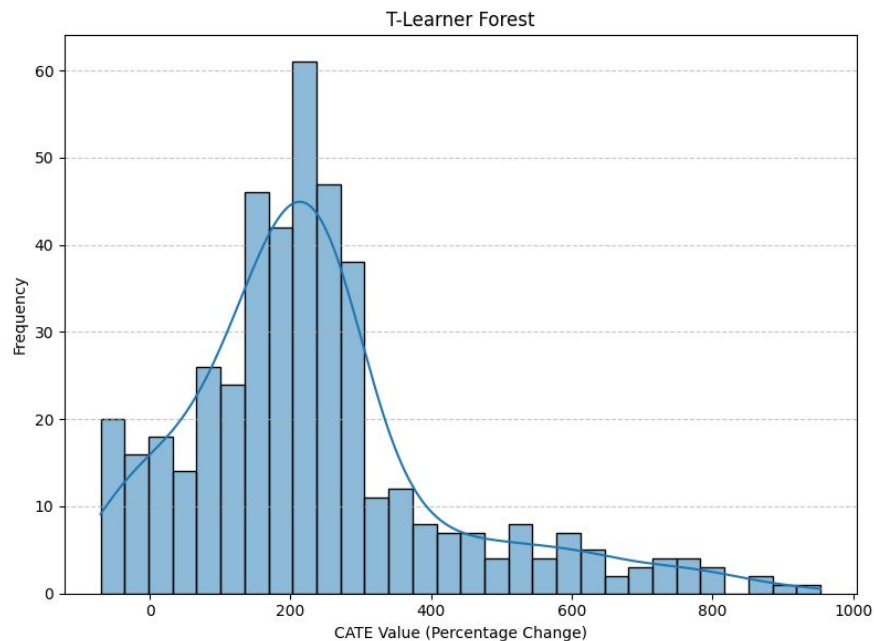
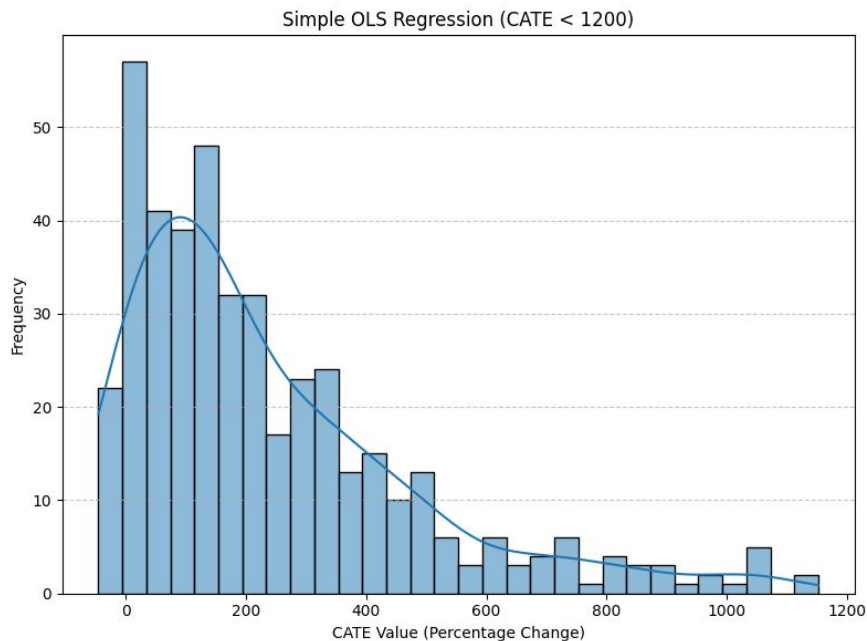
Regression Models:	Mean squared error	Adjusted R-squared
Linear OLS	13.6676	-0.0836
Linear Ridge	13.6407	-0.0815
Linear LAVA	13.6676	-0.0836
Flexible OLS	13.8815	-0.1278
Flexible Ridge	13.5649	-0.1021
Flexible LAVA	13.8815	-0.1278

Linear model: $\log(re_{78}) = \beta_0 + \beta_1 treat + \beta_2 black + \beta_3 educ + \beta_4 age + \beta_5 \log(re_{75}) + \epsilon$

Extracting Treatment Effects - ATE

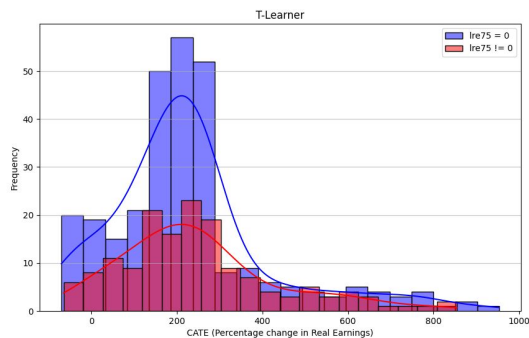
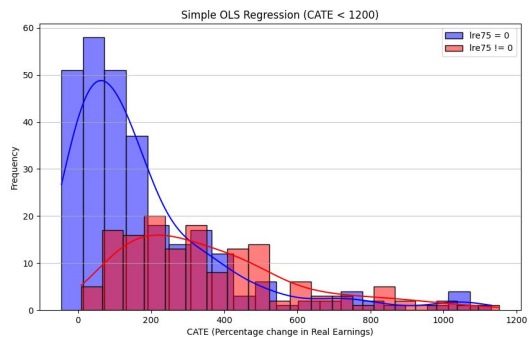


Extracting Treatment Effects - CATE



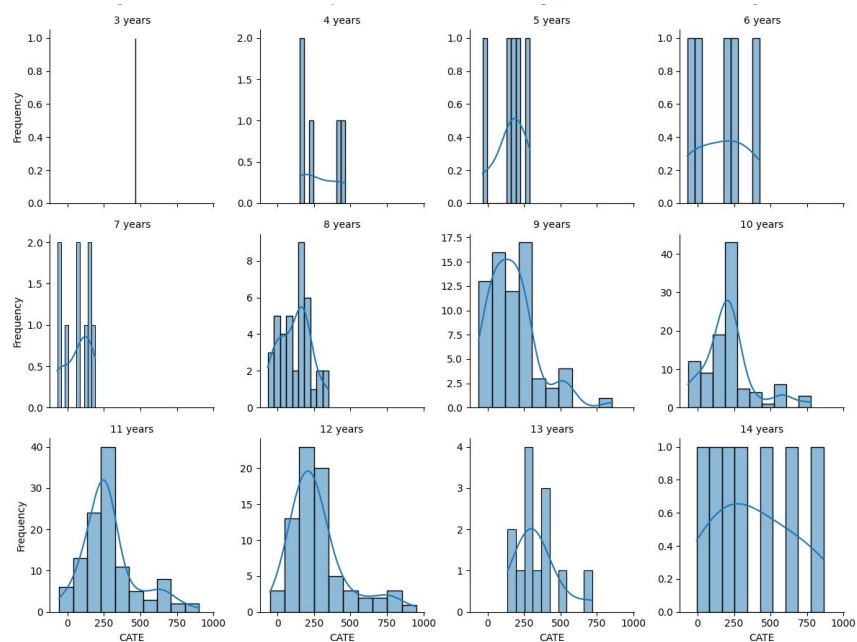
Inspecting Heterogeneity

By Pre-Treatment Income



Inspecting Heterogeneity

By Education (T-Learner CATE < 1200%)



Conclusion

Does participation in a job training program increase real earnings for disadvantaged people?

Which group benefits the most?

Thank you for your attention!